

# Forecasting Exchange Rates using Time Series Analysis

**Objective:** Leverage ARIMA and Exponential Smoothing techniques to forecast future USD to Australian Dollar exchange rates.

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## Part 1: Data Preparation and Exploration

The dataset `exchange_rate.csv` contains daily exchange rate data starting from January 1, 1990.

- **Data Loading:** The `date` column was parsed into a datetime object to set as the index.
  - **Initial Exploration:** The time series was plotted to identify trends. The exchange rate shows significant fluctuations over the decades, with no clear global seasonality but visible local trends.
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## Part 2: The Complete Codebase

This code implements all three requested parts: Data Exploration, ARIMA modeling, and Exponential Smoothing.

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Python
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.holtwinters import ExponentialSmoothing
from sklearn.metrics import mean_absolute_error, mean_squared_error
import numpy as np

# Part 1: Data Preparation
df = pd.read_csv('exchange_rate.csv')
df['date'] = pd.to_datetime(df['date'], dayfirst=True)
df.set_index('date', inplace=True)
series = df['Ex_rate']

# Visualization
series.plot(figsize=(10, 5), title='USD to AUD Exchange Rate (1990-2010)')
plt.show()

# Part 2: Model Building - ARIMA
# Splitting data (80% train, 20% test)
train_size = int(len(series) * 0.8)
train, test = series[0:train_size], series[train_size:len(series)]
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# Fit ARIMA model (Parameters p,d,q selected based on ACF/PACF logic)
arima_model = ARIMA(train, order=(5,1,0))
arima_fit = arima_model.fit()
arima_forecast = arima_fit.forecast(steps=len(test))

# Part 3: Model Building - Exponential Smoothing
# Using Holt's Linear Trend model as rates show a trend but no clear seasonality
exp_model = ExponentialSmoothing(train, trend='add', seasonal=None)
exp_fit = exp_model.fit()
exp_forecast = exp_fit.forecast(len(test))

# Part 4: Evaluation
def evaluate(actual, pred, name):
    mae = mean_absolute_error(actual, pred)
    rmse = np.sqrt(mean_squared_error(actual, pred))
    print(f"{name} -> MAE: {mae:.4f}, RMSE: {rmse:.4f}")

evaluate(test, arima_forecast, "ARIMA")
evaluate(test, exp_forecast, "Exponential Smoothing")

# Visualization of Forecasts
plt.figure(figsize=(12, 6))
plt.plot(test.index, test, label='Actual')
plt.plot(test.index, arima_forecast, label='ARIMA Forecast')
plt.plot(test.index, exp_forecast, label='Exp Smoothing Forecast')
plt.legend()
plt.title('Model Comparison: ARIMA vs Exponential Smoothing')
plt.show()

```

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## Part 3: Evaluation and Discussion

### Model Performance Comparison

Based on the computed metrics (MAE and RMSE), we evaluate the two models:

- **ARIMA:** Effective at capturing the linear relationships in the data. However, it can struggle with long-term forecasts as it tends to revert to a mean or follow a straight trend line if the differencing (d) is not perfectly tuned.
- **Exponential Smoothing:** The Holt's Linear Trend model often performs better on this specific currency dataset because exchange rates tend to follow local trends that are smoothed out more effectively by weighting recent observations more heavily.

### Challenges Encountered

- **Volatility:** Currency exchange rates are highly volatile and influenced by external geopolitical factors not present in the CSV data.
- **Parameter Tuning:** Selecting the optimal  $(p, d, q)$  for ARIMA requires careful inspection of ACF and PACF plots to avoid overfitting.

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## Part 4: Interview Questions and Answers

### Q1: What is the difference between ARIMA and Exponential Smoothing?

**A:** ARIMA is a univariate model that uses past values and errors to predict future ones, focusing on the autocorrelation in the data. Exponential Smoothing (like Holt-Winters) is based on a weighted average of past observations, where weights decay exponentially as the observations get older.

### Q2: Why do we use differencing (the 'd' in ARIMA)?

**A:** Differencing is used to make the time series **stationary** by removing trends or seasonality. Stationarity is a requirement for the ARIMA model to produce reliable statistical forecasts.

### Q3: How do you choose between Simple, Holt's Linear, and Holt-Winters Smoothing?

**A:** Use **Simple Smoothing** if there is no trend or seasonality. Use **Holt's Linear** if there is a trend but no seasonality. Use **Holt-Winters** if the data exhibits both trend and seasonal patterns.

### Q4: What do MAE and RMSE tell us about our forecast?

**A:** MAE (Mean Absolute Error) tells us the average magnitude of the errors in our predictions. RMSE (Root Mean Squared Error) gives a higher penalty to large errors, which is useful if we want to avoid major forecasting misses.